

The Theory of Process

What makes things happen?

by Robert Johansson

Abstract

A discussion of the two process model as a figure/ground gestalt with its roots in Aristotle, Newton, Darwin, Telecommunications, Cybernetics, and Crick and Gamov, and using the Governor as an example. Contrary to mechanistic theory, force is not the only thing that makes things happen. Communication makes things happen.

Aristotle

Aristotelean ontology said that being had two aspects form and substance.

Thus form and substance were causes.

Change of substance was the efficient cause. (You push things around.)

Change of form was the final cause. (The acorn's final cause or purpose was the oak tree.)

Newton

In the 17th Century there arose the scientific dogma of the two causes.

The material cause and the efficient cause.

In Newtonian mechanics: All things stay the same until they are forced to change, and every action has an equal and opposite reaction. Therefore all change can be described by a mathematical equations. There was no final cause. This worked well with stones.

The idea that this is a complete description of all reality is called mechanistic reductionism. Since it is obvious that people do make plans and do make decisions, this is insane. A person that thinks their every day reality is not really real is schizophrenic.

Thus was created the mind/body problem. Which was resolved by dividing reality into two realms: The objective realm of objective reality, and the subjective realm of the mind or imagination which was not really real.

Darwin

In the 19th Century Darwin found that mathematic was insufficient as a description of of the change of form or morphology. It required a more general and abstract understanding. Change in form was a result of choice, or selection. But he retained the intentional fallacy, thus generating the idea of choice without consciousness, choice that was a fundamental characteristic of the universe which he called "natural selection."

Telecommunications

In the 20th Century, the telecommunications people found that radio waves and electronic waves could have form, which they called information. Thus the relationship between mind and body was that matter and energy were the medium of information and communication. Following Darwin they described information as selection within a set, and therefore the final cause was communication which was the translation of choices in one set into choices in another set.

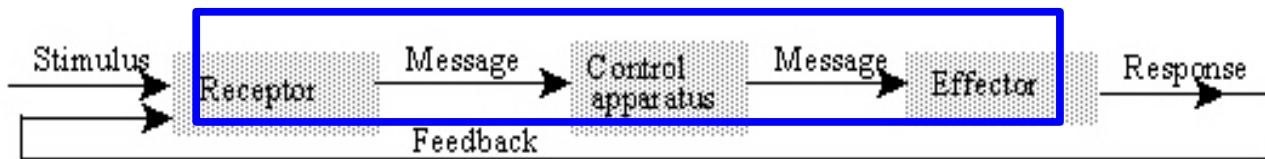
Cybernetics

Later in the 20th Century, the great discovery of cybernetics was that control systems are information systems. Information is neither matter nor energy (You cannot measure it in kilograms or joules.) It is form. Form controls matter and energy. As form it is choice, and thus can function as a command. The final cause was communication. It was feedback, purpose, and adaptation in a command and control system.

Cybernetics and the Two Realm Model

There were many efforts in the cybernetic community to fit the new cybernetics into the old 17th Century two realm model. We will look at three of them.

Von Bertalanffy in 1968 presented the following model.



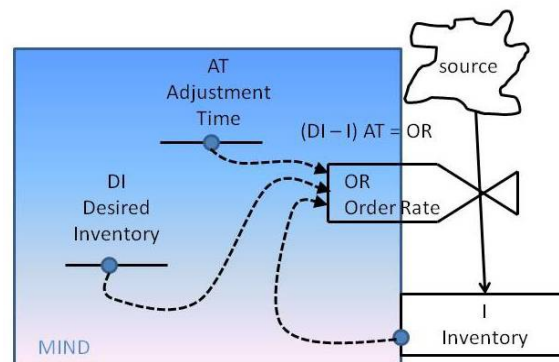
I have added a blue box to indicate the area of mind. Inside the blue box is the subjective world of messages.

Outside the box is the objective world of stimulus and response.

The problem occurs at the transition points. How do you translate information measured in binary digits (bits) into matter and energy measured in kilograms and Joules. They don't translate. They are incommensurate. You cannot describe the one in terms of the other. The Receptor and Effector are supposed to do that, but how? The problem of defining the mind is clear from the position of feedback, is it information or matter/energy?

The most popular "scientific" solution to the mind/body problem is to pretend that it doesn't exist. One way to do that is to pretend that information is "some kind of energy." Another very popular way is to pretend that it is "some kind of probability function." One cyberneticist cheerfully told me "It's all neural nets." That's like saying, "I can explain books. It's all paper." My favourite is "We don't talk about that any more. The smart people have moved on. They talk about complexity."

Jay Wright Forrester's System Dynamics is also a two realm model.

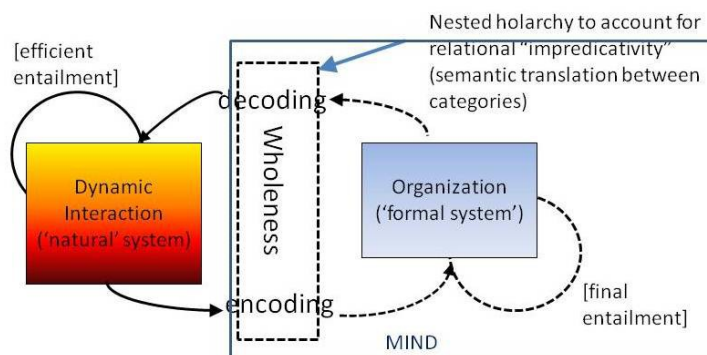


[colour added]

The information processes are the dotted lines, and the material flows are the solid lines. The focus is on material flows (the "real" process). Thus it is essentially linear. It is not good with holarchies. In the World3 model Desired Inventory and Adjustment time had to be plugged in from

historical averages. Thus Limits to Growth was largely ignored, because it did not address the question of how these decisions get made. In mechanistic reductionism there are no decisions. The boundary problem is solve with the valve which is in both realms. The information process controls the form of the valve which controls the flow of material. Information controls form and form controls matter and energy.

John Jay Kineman's model is another two realm model.



[colour added]

Note the dotted lines to indicate the formal system. But you still have the boundary problem. Encoding and decoding are information processes, they translate one kind of information into another kind of information. So they don't solve the boundary problem.

This is finessed by "wholeness" where nested holarchy accounts for "semantic translation between categories." Wholes are a function of information so it still doesn't account for how you get kilograms or Joules out of information or vice versa.

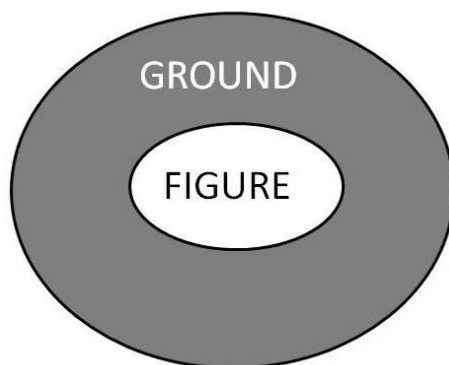
In 1956 **Francis Crick and George Gamov** came up with what they called *the central dogma of microbiology* which presents the two process model or process dualism.

There are two processes inside a cell: the energy processes (which we have been calling body) and the information processes (which we have been calling mind) The important one is the information process. Thus DNA as coded genetic information, which controls the development of the organism. And the medium is DNA, an acid: a molecule with energy. The important one is the information process.

Gestalt

The basic form of gestalt is **the figure ground gestalt**.

We can perceive the word "figure" because it is black against a white ground.



We can perceive the word "ground" because it is white against a black ground.

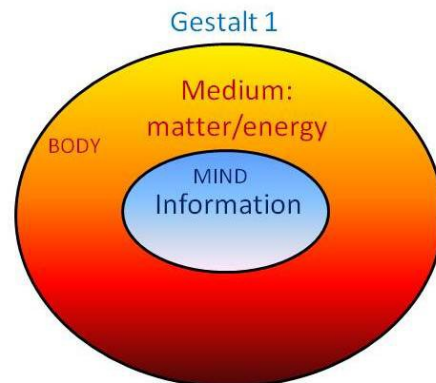
We perceive the meaning of the figure only in the context of the ground. Because we perceive the figure we don't perceive the ground. Let me repeat that. Because we perceive the figure we don't

perceive the ground. The ground is the whole that allows us to perceive the figure.

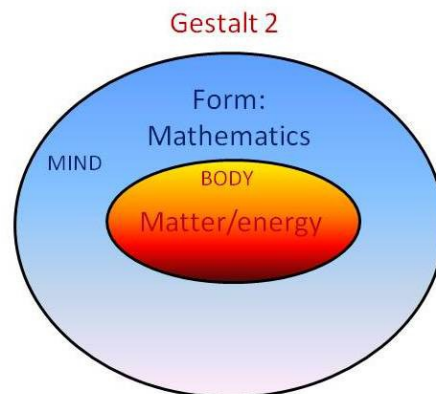
How can you have a reality with two different processes? The answer is that reality is a figure/ground gestalt. The basic principle of holism is that it is the whole that gives meaning to the part. The meaning of part number 782,394 is that it is part of an Airbus. Without the Airbus it is just junk. You have all seen a debris field. You do not get a whole by adding up a bunch of things. You are not more holistic if you add more parts. You are holistic if the whole gives meaning to the parts.

Thus there are two ways of looking at reality.

Gestalt One is our old friend telecommunications: Information and communication are perceived against the background, the media of matter and energy.

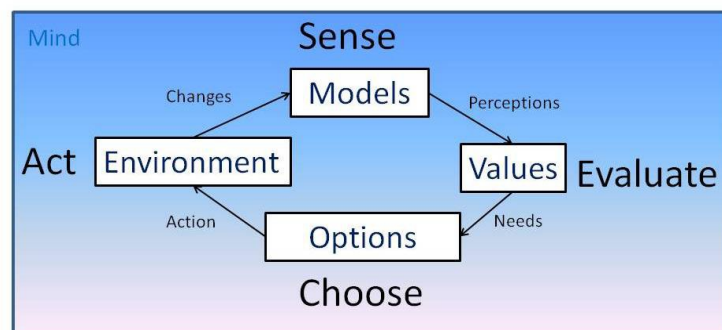


Gestalt Two is our old friend cybernetics. Matter and energy are perceived against the background of form. Information is the whole that controls matter and energy.



The Cybernetic Control Loop

Why is that important? Because a cybernetic control loop is an information process.



So let's look at an example: The greek word “cybernetes” means governor. So let's look at the governor.

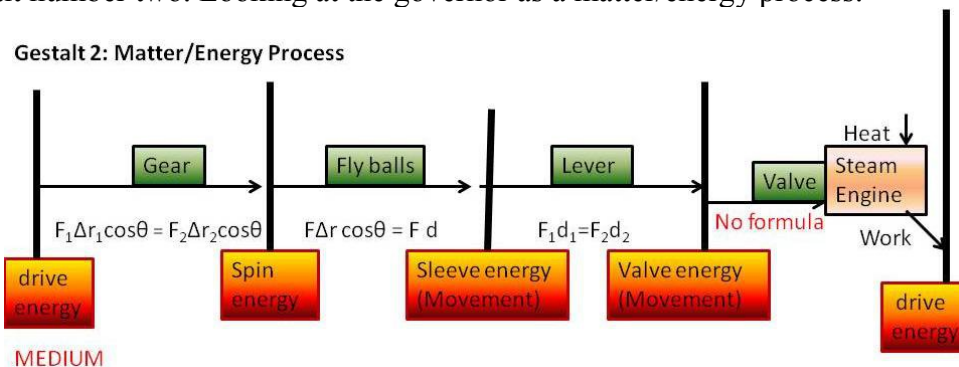
The governor needs to **sense** what is going on. So the gear translates the changes in drive speed into changes in spindle speed. At the meta-level this is a process of modelling. In this case the spindle is a scale model of the drive shaft.

The governor then needs to **evaluate** what is going on. The fly balls translate the change in spin into changes in the position of the sleeve. If the spindle goes fast the sleeve goes up, and if the spindle goes slower the sleeve goes down.

The governor then needs to **choose** from the available options. With the lever there are two options called negative and positive feedback. Positive feedback is when the lever translates a rise in the sleeve into an increase in the valve opening. Negative feedback is when the lever translates the rise in the sleeve into a decrease in the valve opening. The governor is a homeostatic system. Its goal is to maintain the drive speed at a constant level, thus negative feedback.

Finally the **acting** to cause change in the environment. The steam engine is the environment. The valve opening controls the flow of steam to the steam engine and thus controls the change in the speed of the drive shaft. Which is where we began. Thus a cybernetic loop.

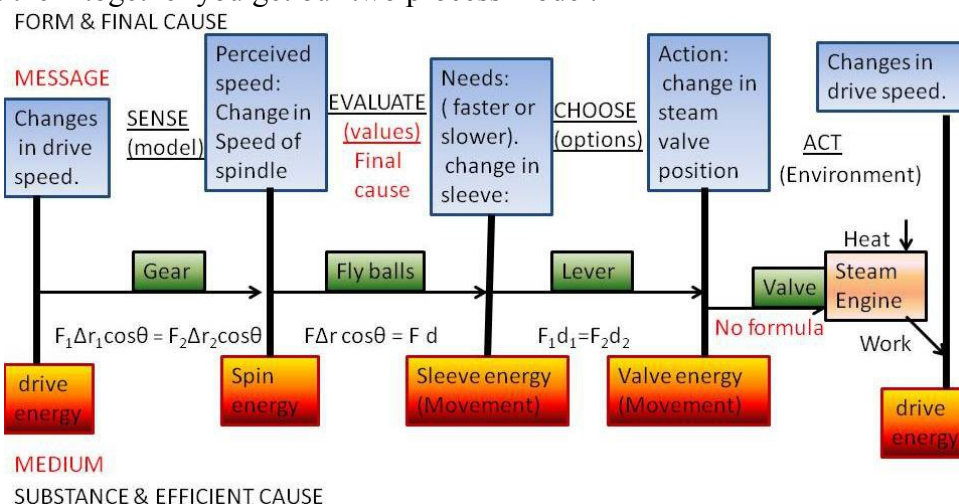
Now for gestalt number two. Looking at the governor as a matter/energy process.



The energy process is an equation: one kind of energy equals another kind of energy.

Thus the drive energy equals the spindle energy equals the movement of the sleeve equals the movement of the valve. But there is no equation of the final action of the governor. The valve operates at the meta or control level to change the form of the steam engine. The information process controls the form of the valve which controls the flow of material. Form controls matter and energy.

When you put them together you get our two process model.



The matter and energy process of the material and efficient cause, are the media for the formal and final

causes, the information and communication. There is one small adjustment to the Aristotelean terms. The final cause or purpose of the process is found the the core values of the cybernetic process.

Force

We were all taught the mechanistic understanding of change that everything remains the same until it is forced to change. As a result we believe that all change is the result of force and we look for the forces that caused things to happen or we look for some force that we can use to make change happen. When I hear someone say “The reality is . . . “ I know that they have bought into the idea that only force makes things happen.

Communication

What we learn from the study of process is that communication makes things happen. Our behaviour is controlled by our decisions, and our decisions are controlled by our ideas, and our ideas are the result of communication.

You cannot force someone to change their mind. To change their mind they have to make a decision, they have to make a choice. Power is the ability to make a decision, and we all have that power.

But we can communicate with people. Communication begins with listening to people, so that we learn their language. Then we can share. Love makes change happen.